Exponential functions

1

a 5 **b** 80 **c** $\frac{5}{4}$ **d** 320

2

a 1 **b** $\frac{9}{16}$

3 $50\,313.28$

4

a 125 **b** 1 **c** $\frac{1}{5}$ **d** -2

5

a $(5, -32)$ **b** $\left(-3, -\frac{1}{8}\right)$ **c** $\left(-1, -\frac{1}{2}\right)$ **d** $(4, -16)$

6 $m = 8$

7

a $n = 8$ **b** $k = 4$ **c** $m = -2$

Characteristics of exponential functions

8 No, both functions are always greater than zero.

9 None, for all values of x it will have a positive value.

10

a $(0, 1)$ **b** $(0, 1)$ **c** $(0, -1)$ **d** $(0, -1)$

11

a 1 **b** 1

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12

a $\frac{1}{81}$

b 1

c 81

d It gets larger, approaching infinity.

e It gets smaller, approaching zero.

13

a Positive

b 0

c $y = 0$ is a horizontal asymptote of the curve.

14 There is no maximum function value. As x decreases, y approaches 0 but never quite gets there.

15

a $y = 4^x$

b y is increasing at an increasing rate.

16 $y = 5^x$

17 $y = 5^{-x}$

18

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	$59\,049$

b As x increases, the function increases at a faster and faster rate.

c All real x

d $y > 0$

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19

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

b As x increases, the function decreases at a slower and slower rate.

c 1

d All real x

e $y > 0$

20

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32	-1024

b No. For $a > 0$ the expression a^x is greater than zero for all real x. Now, $-(a^x)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.

e All real x

f $y < 0$

21

a $(0, 1)$ **b** No **c** All real x **d** $y > 0$

e $y = 128$ **f** $x = 8$

22

a Example answers:

- None of the curves cross the x-axis.
- They all have the same y-intercept.
- As x gets large, the function values approach 0.

b 1

c As x gets large, the function values approach 0.

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23

a $(0, 1)$

b As x gets large $y = 3^x$ becomes large while $y = 3^{-x}$ approaches 0.

24

a No **b** Yes **c** Yes **d** No

Graphs of exponential functions

25

a Yes **b** No **c** No **d** No

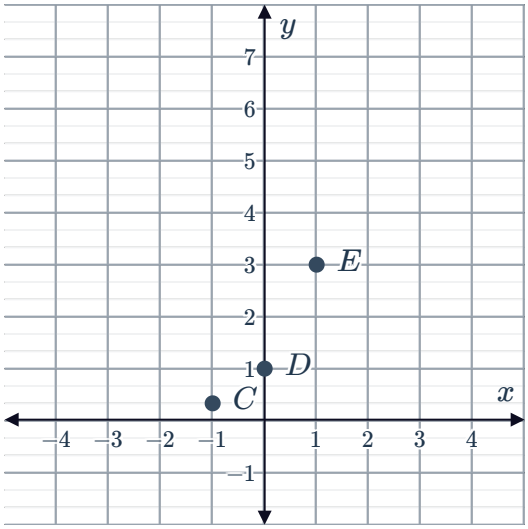
e No **f** No **g** No **h** Yes

26

a

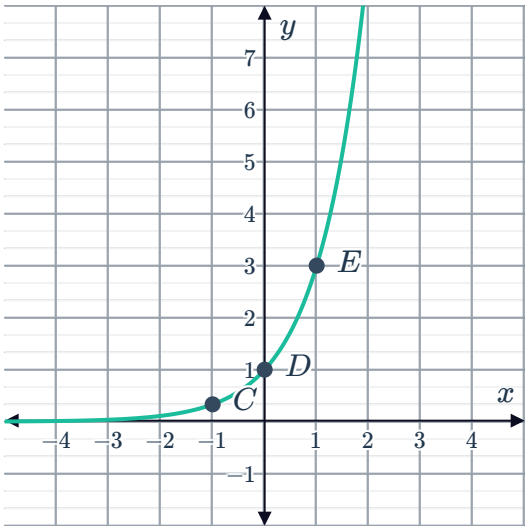
Point	Coordinates
A	$\left(-3, \frac{1}{27}\right)$
B	$\left(-2, \frac{1}{9}\right)$
C	$\left(-1, \frac{1}{3}\right)$
D	$(0, 1)$
E	$(1, 3)$
F	$(2, 9)$
G	$(3, 27)$

b



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c

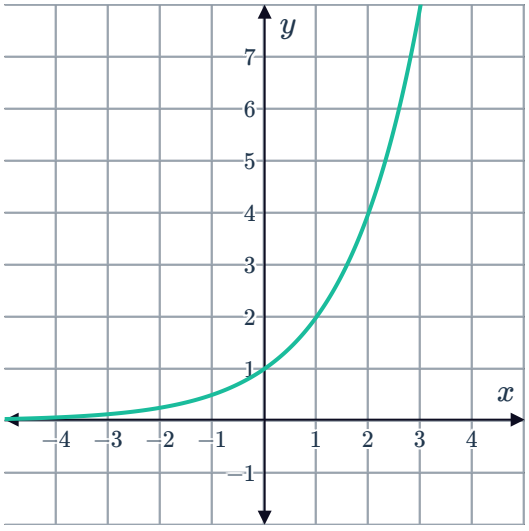


27

a

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

b



c Increases

d Decreases

e The curve never crosses the x-axis.

f 1

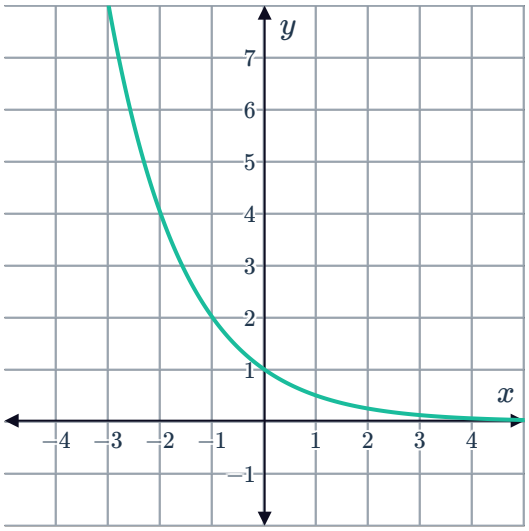
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28

a

x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

b



c Decreases

d Increases

e The curve never crosses the x-axis.

f 1

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29

a

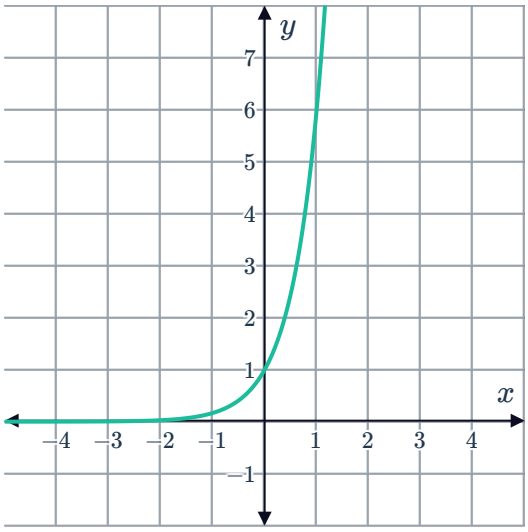
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{216}$	$\frac{1}{36}$	$\frac{1}{6}$	1	6	36	216

iii $y = 0$

iv



b

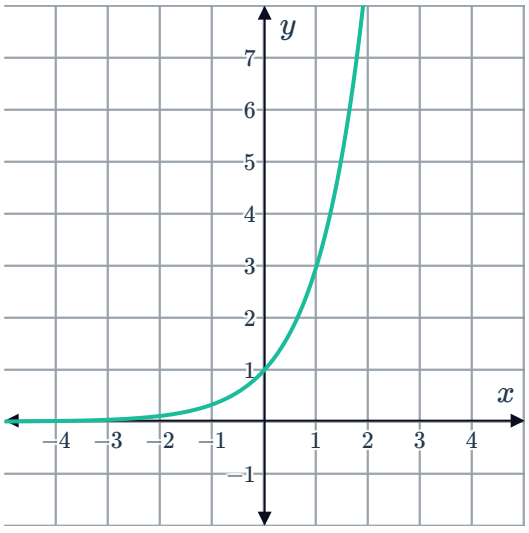
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

iii $y = 0$

iv



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c

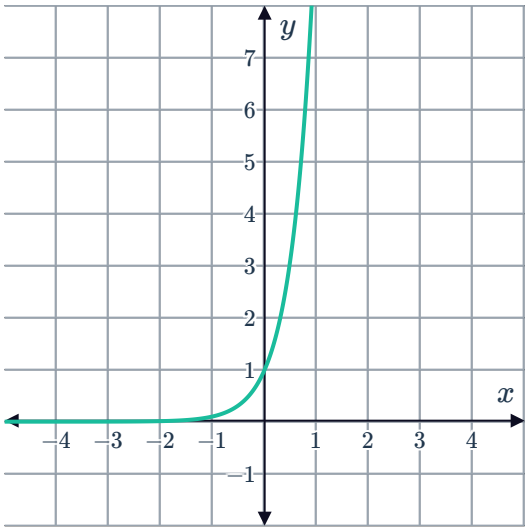
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{1000}$	$\frac{1}{100}$	$\frac{1}{10}$	1	10	100	1000

iii $y = 0$

iv



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d

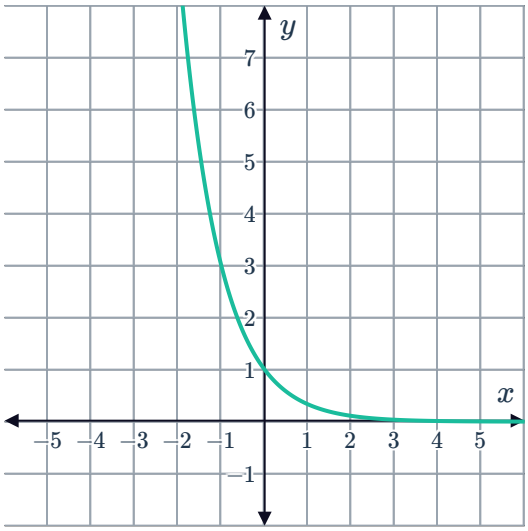
i $(0, 1)$

ii

x	-3	-2	-1	0	1	2	3
y	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

iii $y = 0$

iv



30

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

b ∞

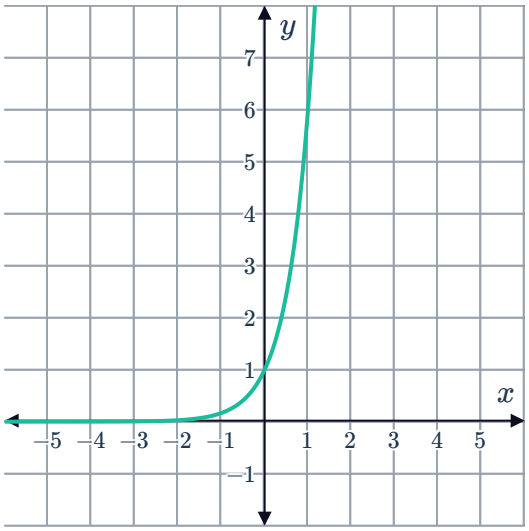
c 0

d 1

e 0

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f



31

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

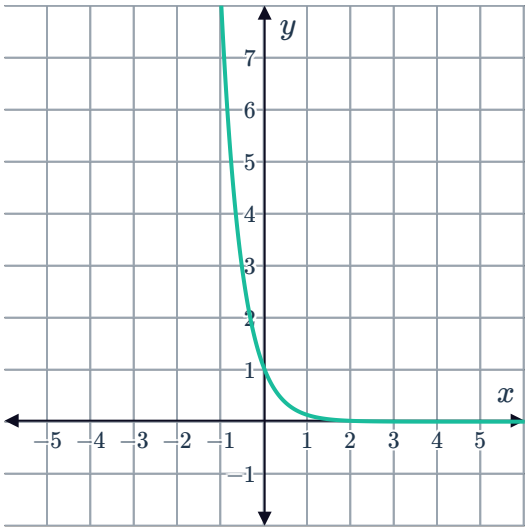
b 0

c ∞

d 1

e 0

f



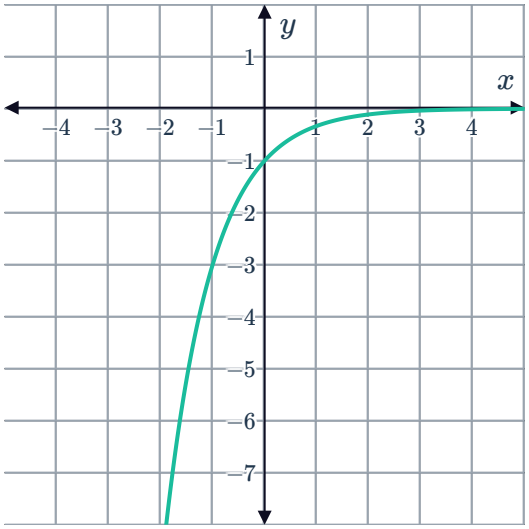
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34

a $(0, -1)$

b $y = 0$

c



d Increasing function

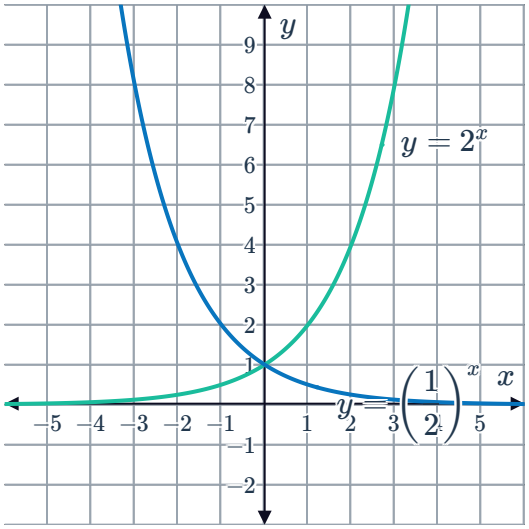
35

a

i Yes **ii** Yes **iii** No **iv** No

b Reflection across the y-axis.

c



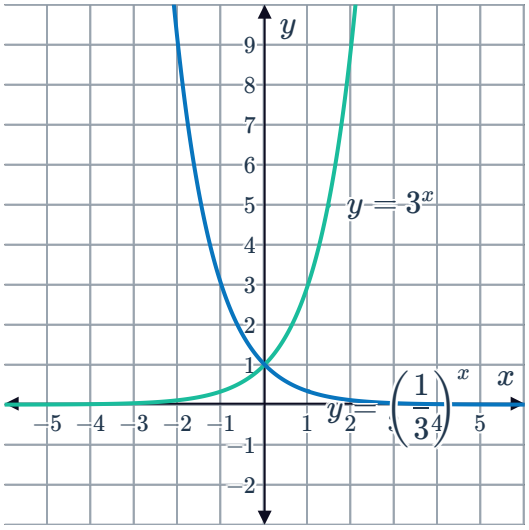
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36

a $y = 3^{-x}$

b Reflection about the y-axis.

c



37

a $y = 0$

b $y = 0$

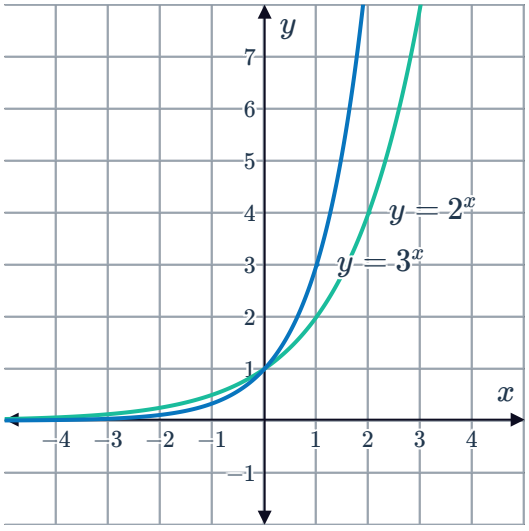
c

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

d

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

e

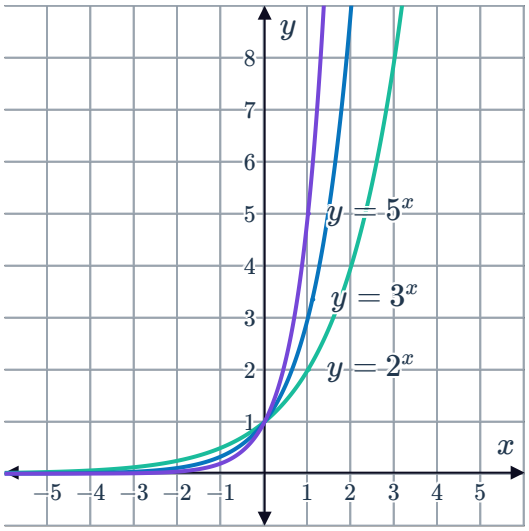


f $(0, 1)$

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38

a



b Shared y-intercept of $(0, 1)$

c As x gets larger, the function increases at an increasing rate and the values approach ∞.

39

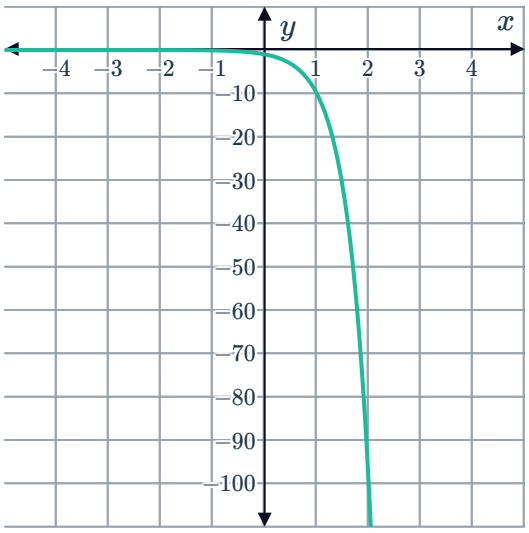
a Graph R

b For $x < 0$, the graph of $y = 6^x$ is below the graph of $y = 4^x$. This is because negative values of x result in fractional function values, and for any negative value of x, 6^x will result in a smaller value than 4^x.

40

a Yes. For any value of x, 10^x is always positive. Therefore, -10^x is always negative.

b



c No solution

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